

ASG AIRLINES – DATA ANALYTICS PROJECT

Data Engineering and Power BI Dashboard

Project Documentation

Prepared as part of the Data Engineering / Analytics Case Study

Data Source	Airline Excel dataset
Main Tools	Python, Pandas, Power BI
Main Output	Clean data, KPI files and Power BI dashboard
Dashboard Pages	Executive Overview, Flight Analysis, Bookings & Revenue, Data Quality

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1. Project Overview

The aim of this project is to take airline data from the raw Excel file, check and clean it, prepare useful tables and finally present the results in Power BI. The main focus is on flights, bookings, payments, routes and data quality.

The pipeline is divided into simple stages: ingestion, validation, cleaning, modelling, KPI calculation and reporting. This makes the process easier to follow and also makes it possible to check problems before the data reaches the dashboard.

2. Data and Processing

The project starts with the airline Excel workbook. The required sheets are loaded using Python and Pandas. Basic checks are then applied to find missing values, duplicates, invalid categories, incorrect numbers, timestamp problems and reference mismatches.

During cleaning, text fields are trimmed and missing values are handled. Duplicate records are identified and recorded. Booking statuses are standardised as CONFIRMED, CANCELLED, PENDING or INVALID/UNKNOWN. Invalid payment amounts are separated so that they do not affect valid payment analysis.

Passenger information is treated carefully. Aadhaar numbers, phone numbers and email addresses are masked in the cleaned data. This keeps the data useful for analysis without exposing the original personal details.

3. Flight Duration and Overnight Handling

Flight duration is calculated from the departure and arrival timestamps. The difference between the two timestamps is converted into minutes. If the arrival date is on the next day, the flight is marked as an overnight flight. This prevents an overnight journey from being treated as a negative or very small duration.

Very unusual durations are flagged as outliers. Multi-leg or ambiguous flight references are not forced into an incorrect match. Instead, they are kept as unresolved records so the original data is not silently changed.

The final model contains flight, booking, airline, route and passenger information. The main fact tables are Fact_Flights and Fact_Bookings, with dimension tables for Airline, Route, Flight and Passenger.

Table	Purpose
Fact_Flights	Flight-level records and duration information
Fact_Bookings	Booking status and booking-level analysis
Dim_Airline	Airline names and airline keys
Dim_Route	Route codes and route details
Dim_Flight	Flight identifiers and flight timing
Dim_Passenger	Passenger information with protected identifiers

4. Pipeline and Data Flow

The overall flow used in the project is shown below. Each stage produces data that is used by the next stage.

RAW	CHECK	CLEAN	MODEL	KPI	REPORT
Excel workbook	Validation and quality checks	Cleaning and standardisation	Star-schema tables	Business and quality KPIs	Power BI dashboard

The Gold layer contains the analysis-ready CSV files such as fact_flights, fact_bookings, dim_airline, dim_route, flight_kpis, booking_kpis, payment_kpis, route_flight_traffic, airline_flight_distribution and data quality reports.

5. Power BI Dashboard

The dashboard was created to keep the main results easy to read. Filters are available for airline, booking status and route where they are useful. KPI cards give a quick view of the overall numbers, while charts are used for comparison.

The dashboard contains four pages: Executive Overview, Flight Analysis, Bookings & Revenue, and Data Quality.

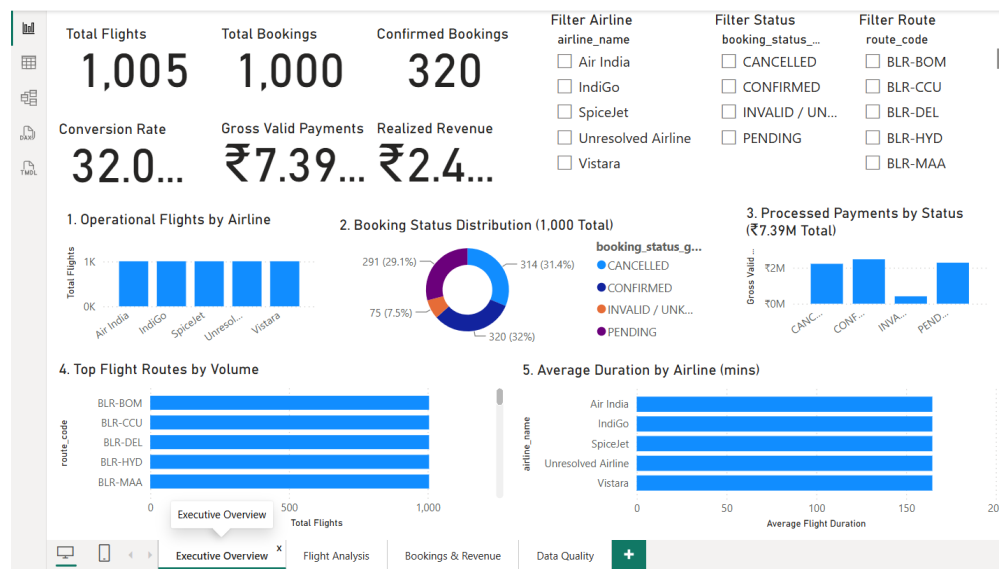


Figure 1. Executive Overview

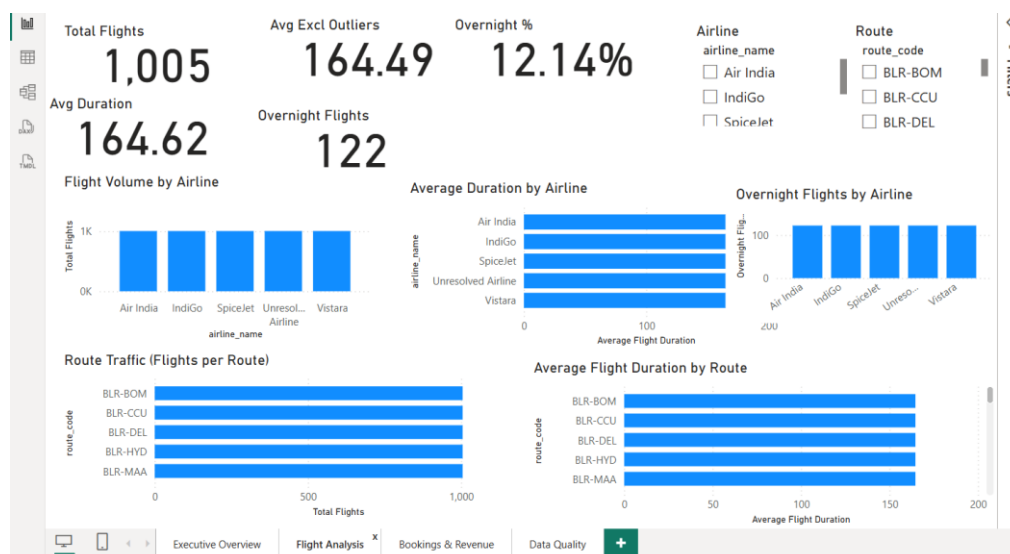


Figure 2. Flight Analysis

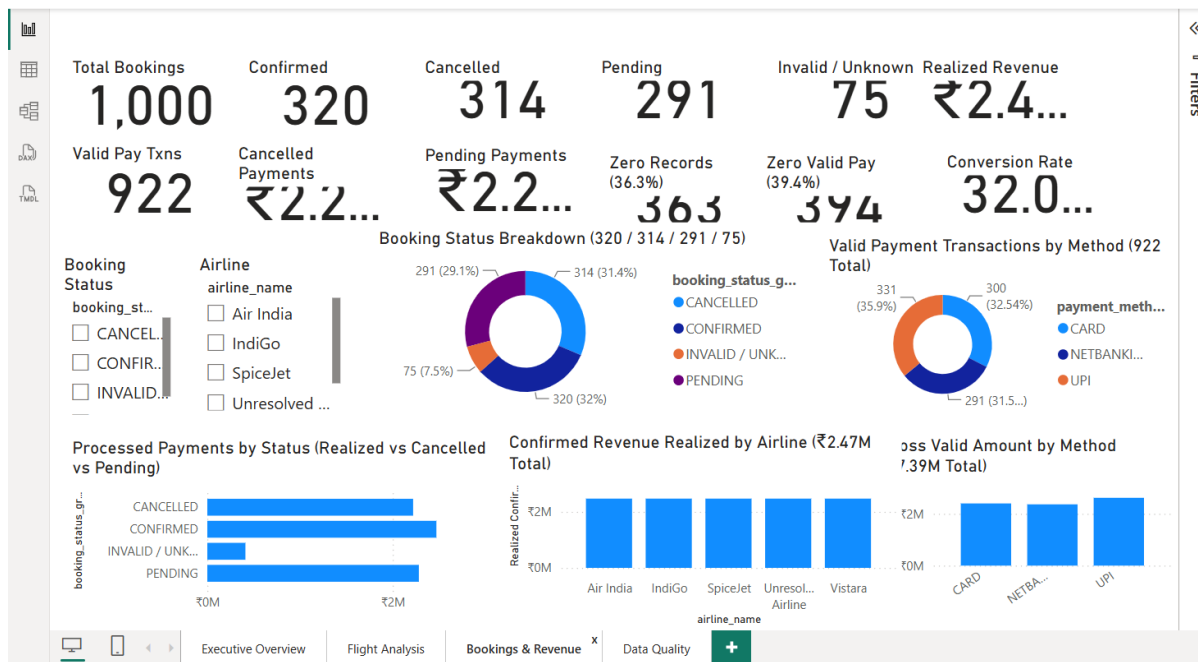


Figure 3. Bookings and Revenue

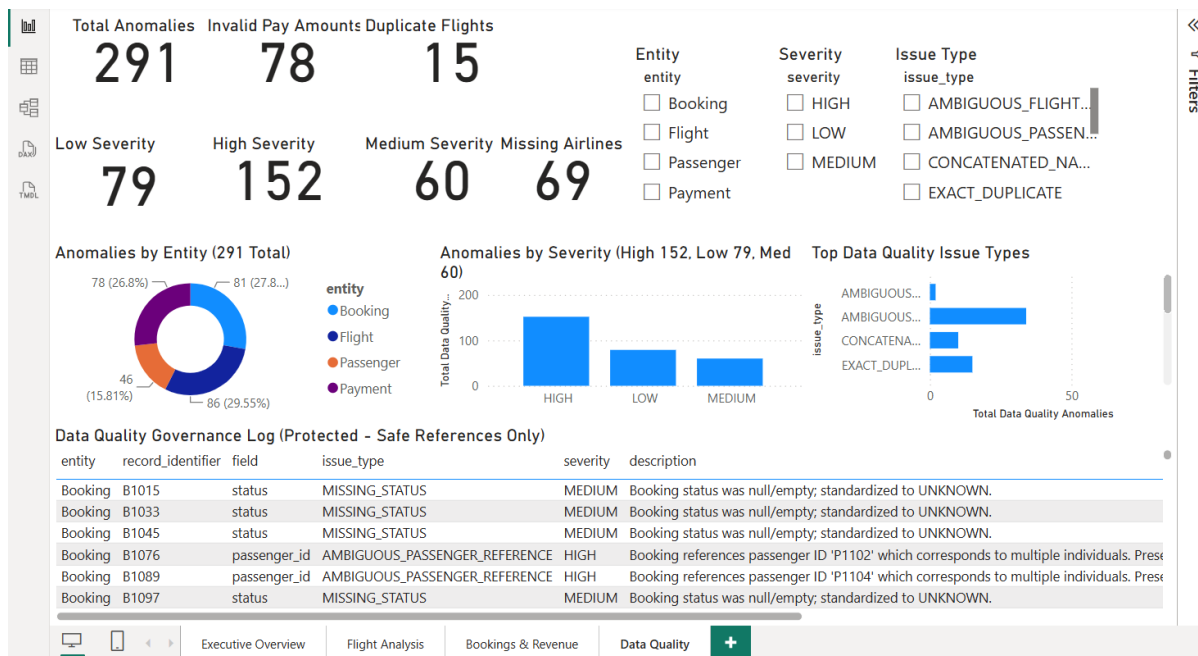


Figure 4. Data Quality

6. Main KPIs and Data Quality

Some of the main values shown in the dashboard are 1,005 total flights, 1,000 total bookings, 320 confirmed bookings and 122 overnight flights. The flight analysis page shows an average duration of about 164.62 minutes and an overnight percentage of 12.14%.

For bookings and payments, the report shows confirmed, cancelled, pending and invalid/unknown bookings separately. This helps avoid treating every booking as completed revenue.

The Data Quality page reports 291 total anomalies, including 152 high-severity, 79 low-severity and 60 medium-severity issues. It also shows invalid payment amounts, duplicate flights, missing airlines and other issue types. These records are kept in a governance log for review.

7. Assumptions and Final Output

The project keeps unresolved references instead of guessing the correct record. An unknown booking status is kept as UNKNOWN. Flight durations are based on the cleaned departure and arrival timestamps. Confirmed booking payments are treated separately from cancelled, pending and invalid payments.

The final project contains the Python pipeline, raw input, cleaned/Gold CSV files, KPI and data quality outputs, and the Power BI project. The Power BI report is intended to give a simple business view of flight operations, bookings, revenue and data quality.

Overall, the project follows the complete flow from raw data to a usable dashboard. The main purpose is not only to show charts, but also to show how the data was checked, cleaned and prepared before reporting.

8. Submission Checklist

- Raw airline Excel data included.
- Python ingestion, validation, cleaning and modelling files included.
- Gold analysis-ready CSV files included.
- Flight, booking, payment and data-quality KPIs included.
- Power BI dashboard project included with four report pages.
- Dashboard screenshots included in this document.
- Cleaning, duration, overnight and PII handling explained.
- Data model, data flow, assumptions and final outputs documented.